Task 1: Introduction to Artificial Intelligence

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is the branch of computer science that builds machines and software able to do tasks that normally need human intelligence, such as learning, reasoning, problem-solving, understanding language, and recognizing images. Instead of following only fixed instructions, an AI system can learn from data and improve with experience. The term was introduced by John McCarthy in 1956.

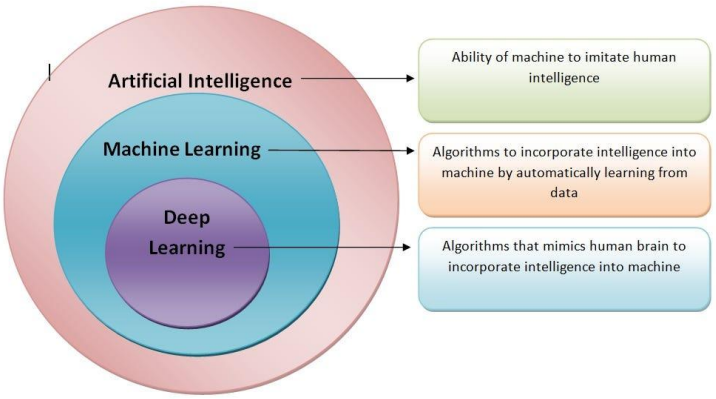
2. Types of AI

Type	Full Name	Description	Example
ANI	Artificial Narrow Intelligence	Performs one specific task well. All AI today is ANI.	Siri, Google Maps, spam filters
AGI	Artificial General Intelligence	Would match human intelligence across any task. It does not exist yet.	Theoretical
ASI	Artificial Super Intelligence	Would surpass human intelligence in every area.	Theoretical

3. AI vs Machine Learning vs Deep Learning

- **AI** is the broad goal of making machines act intelligently.
- **Machine Learning (ML)** is a subset of AI where machines learn patterns from data instead of being explicitly programmed. Example: an email spam filter.
- **Deep Learning (DL)** is a subset of ML that uses multi-layered neural networks, inspired by the human brain. It needs large data and computing power. Example: face recognition, ChatGPT.

Think of it as three nested circles: AI contains ML, and ML contains DL.



4. Real-Life Applications

- **Healthcare:** detecting diseases in X-rays and scans
- **Education:** personalized learning apps
- **Entertainment:** Netflix and YouTube recommendations
- **Transport:** self-driving cars, Google Maps traffic prediction
- **Banking:** fraud detection
- **Smartphones:** voice assistants (Siri, Google Assistant), face unlock
- **E-commerce:** product suggestions on Amazon

5. Advantages and Limitations

Advantages

- Works 24/7 without getting tired
- Fast and accurate on repetitive tasks
- Handles huge amounts of data
- Reduces human error in risky tasks

Limitations

- Expensive to build and maintain
- Needs large amounts of quality data
- Can be biased if the training data is biased
- Lacks real creativity and emotions
- May replace some jobs and raises privacy concerns

Task 2: Setting Up AI Development Environment

A. Install and configure: Python

To install Python on Windows, download the official installer from Python Downloads or use the built-in Python Install Manager from the Microsoft Store. [1, 2]

Step-by-Step Installation

1. Download the Installer:
    - Go to the Python Windows Releases page and download the 64-bit Windows installer for the latest version (such as Python 3.15).
    - Alternatively, open PowerShell or Command Prompt and install via the Windows Package Manager: [1, 2]

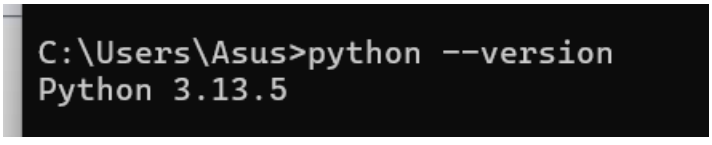
Code: winget install Python.Python.3 --scope machine

2. Run the Installer:
    - Double-click the downloaded .exe file.
    - Important: Check the box at the bottom that says "Add Python to PATH" (or Add python.exe to PATH) so you can run Python commands from your terminal.
    - Click Install Now for a standard setup, or choose Customize installation if you want to change the destination folder or select specific features like pip and IDLE. [1, 2, 3]
3. Verify the Installation:
    - Open Command Prompt or PowerShell from the Start menu.
    - Type the following command to check your version:

Code: python --version

    - Or use the Python launcher: [1, 2, 3, 4]

Code: py –version

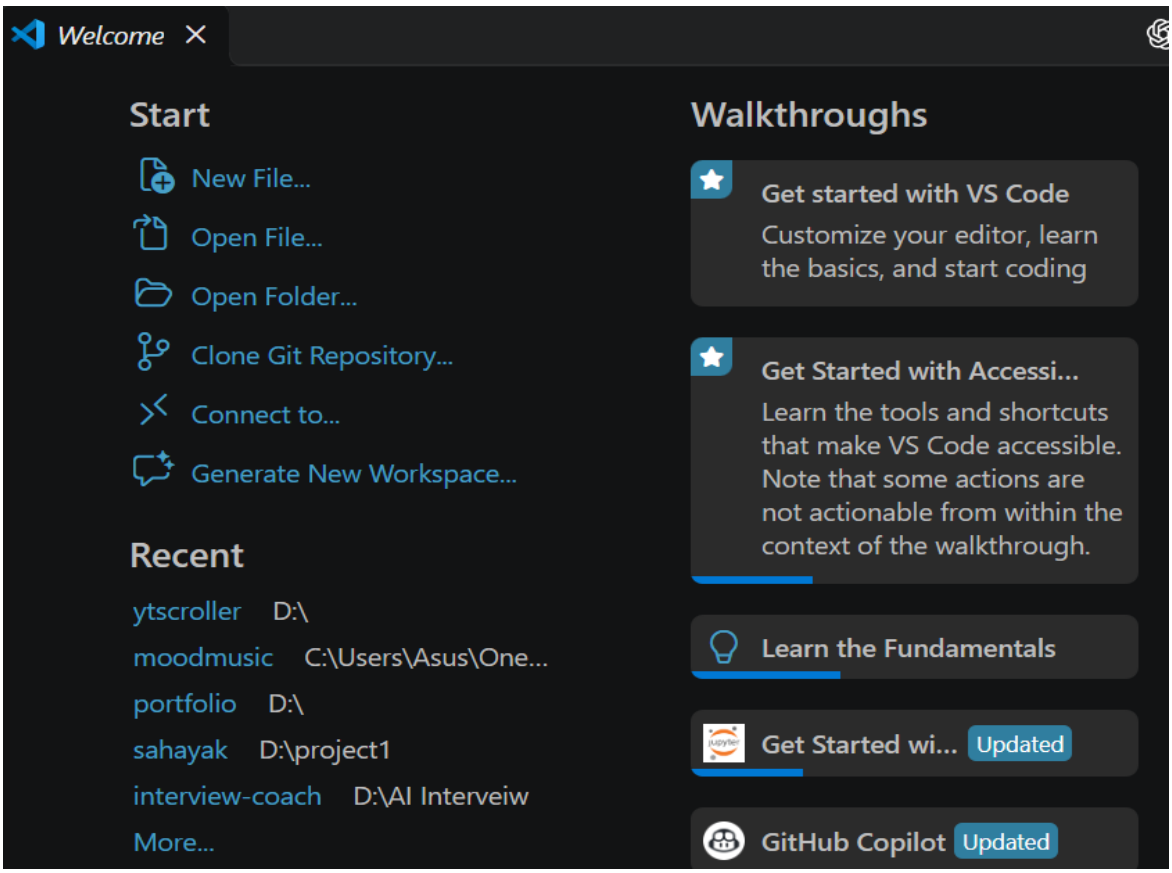


B. VS Code

To install Visual Studio Code on Windows 10 or 11, download the official installer from the Visual Studio Code Download Page. [1, 2]

Installation Steps

- **Download the Installer:** Go to the Visual Studio Code Download Page and click the **Windows** button (User Installer x64 is recommended for most users as it does not require administrator rights). [1, 2, 3]
- **Run the Executable:** Open your Downloads folder and double-click the installer file (e.g., VSCodeUserSetup-x64-{version}.exe). [1, 2]
- **Accept the Agreement:** Select **I accept the agreement** and click **Next**. [1, 2]
- **Choose Options:** Keep the default installation location and Start Menu folder, or check additional tasks such as creating a desktop icon and adding "Open with Code" to the Windows Explorer context menu. [1, 2, 3]
- **Install and Launch:** Click **Install**, wait for the setup to finish, and click **Finish** to launch Visual Studio Code.



C. Jupyter Notebook

Installing **Jupyter Notebook** can be done in two primary ways depending on your needs: using Anaconda (best for data science and beginners) or using pip (best if you already have Python installed and want a lightweight setup). [1]

Here are the step-by-step instructions for both methods.

Method 1: The Anaconda Setup (Recommended for Data Science)

This is the easiest option because it automatically bundles Python, Jupyter Notebook, and popular data analysis libraries like Pandas and Matplotlib. [1]

1. Go to the Anaconda Download Page and download the installer for Windows. [1, 2]
2. Run the downloaded .exe installer. Follow the setup wizard, accepting the default options ("Just Me" or "All Users" work fine). [1, 2]
3. Once finished, open your Windows **Start Menu**, search for **Anaconda Prompt**, and click to launch it. [1, 2]
4. In the prompt window, type the following command and press **Enter**:

Code: jupyter notebook

5. A command window will run, and the Jupyter Notebook dashboard will automatically pop up in your default web browser. [1, 2]

Method 2: The pip Setup (Lightweight Approach)

If you already have Python configured and prefer not to download the large Anaconda package, you can install it directly through the Python package manager. [1, 2]

1. Open your Windows **Command Prompt** (press Win + R, type cmd, and hit Enter).
2. Verify that you have Python and pip running correctly by entering:

Code: python --version

3. Update your package manager to prevent installation conflicts:

Code:python -m pip install --upgrade pip

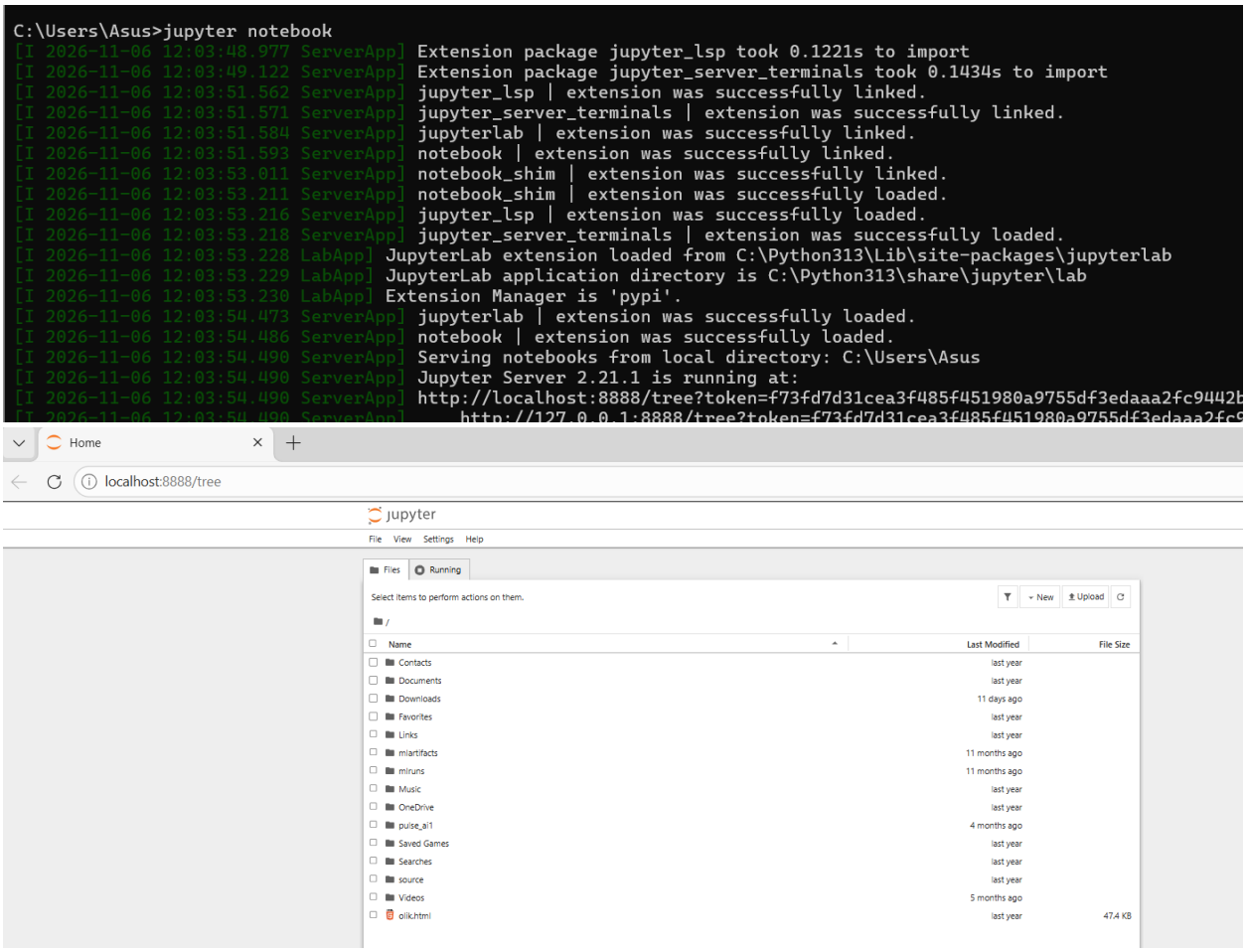
Use code with caution.

4. Run the installation command for the classic Jupyter Notebook tool:

Code: pip install notebook

5. Launch your environment by running: [1, 2, 3, 4]

Code: jupyter notebook



D. Required libraries (NumPy, Pandas)

```
[1]: import numpy as np
     import pandas as pd

     print("NumPy version:", np.__version__)
     print("Pandas version:", pd.__version__)
```

```
NumPy version: 2.3.2
Pandas version: 2.3.1
```

E. Write a simple Python program that prints:

```
[2]: print("Welcome to Artificial Intelligence")
```

```
Welcome to Artificial Intelligence
```

Task 3: Python for AI

Write programs for: Variables Data Types User Input Operators

[3]:
```python
# Variables and data types
student_name = "Rahul"      # str
marks = 72                  # int
attendance = 85.5           # float
is_enrolled = True          # bool
print(type(student_name), type(marks), type(attendance), type(is_enrolled))

# User input + operators: pass/fail predictor
name = input("Enter student name: ")
marks = int(input("Enter marks (0-100): "))

if marks >= 40:
    print(name, "is predicted to PASS")
else:
    print(name, "is predicted to FAIL")

# Temperature conversion for a weather AI
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9 / 5) + 32
print("Temperature in Fahrenheit:", fahrenheit)

# Age eligibility checker
age = int(input("Enter your age: "))
print("Eligible to vote" if age >= 18 else "Not eligible to vote")

# Operators demo
a, b = 10, 3
print(a + b, a - b, a * b, a / b, a // b, a % b, a ** b)
print(a > b, a == b, a > 5 and b < 5)
```

```
<class 'str'> <class 'int'> <class 'float'> <class 'bool'>
Enter student name:  Rupa
Enter marks (0-100):  60
Rupa is predicted to PASS
Enter temperature in Celsius:  76
Temperature in Fahrenheit: 168.8
Enter your age:  25
Eligible to vote
13 7 30 3.3333333333333335 3 1 1000
True False True
```

Task 4: AI Case Study

Netflix Recommendation System

Introduction

Netflix is a streaming platform with millions of users across the world. With thousands of movies and shows available, finding something to watch can be difficult. To solve this, Netflix uses an AI-based recommendation system that suggests content matched to each user's taste. Most of what people watch on Netflix comes from these recommendations.

How AI is Used

Netflix uses machine learning algorithms to study user behaviour and predict what a person will enjoy. Every user sees a different home screen, even the thumbnails and row order are personalized. The system mainly uses collaborative filtering (recommending what similar users liked) and content-based filtering (recommending titles similar to what you already watched).

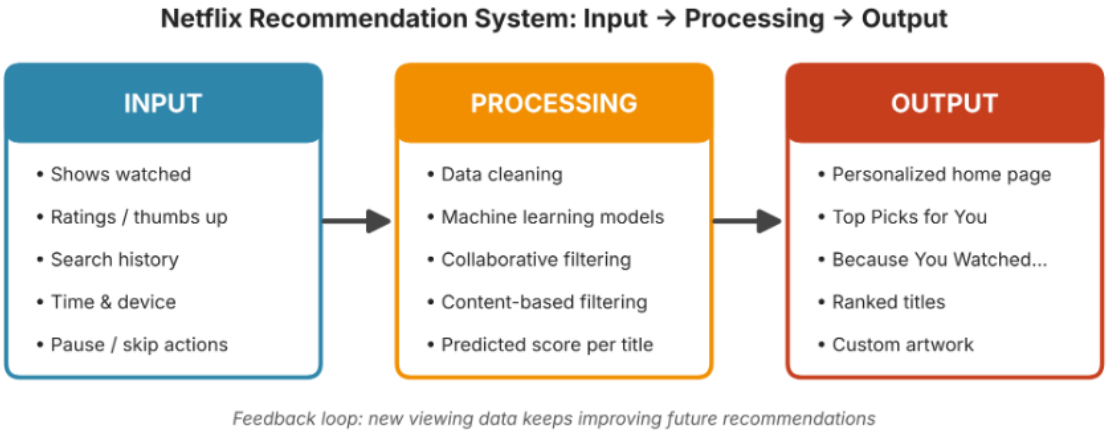
Input

The system collects data such as:

- Shows and movies watched, and how long they were watched
- Ratings and thumbs up or down
- Search history
- Time of day and device used
- Whether a user paused, rewound, or skipped
- Genre, actors, and language of each title

Processing

The collected data is cleaned and fed into machine learning models. The models find patterns, for example that a user who watches crime thrillers at night also tends to like detective series. The system groups users with similar tastes and compares them. It then gives each title a predicted score for each user. These models keep learning and updating as the user watches more content.



Output

The output is a personalized home page with rows like “Top Picks for You,” “Because You Watched…,” and “Trending Now.” Titles are ranked from most to least relevant, and artwork is chosen to suit the user’s interests.

Benefits

- **For users:** saves time and gives relevant suggestions, so they discover new shows
- **For Netflix:** keeps viewers engaged, reduces cancellations, and saves money by helping retain subscribers
- **Personalization:** every user gets a unique experience
- **Scalability:** works for millions of users at once

Conclusion

Netflix’s recommendation system is a strong example of how AI turns raw data into useful, personalized experiences. As more data is collected, it becomes more accurate over time.

Task 5: Mini AI Project

Build a simple Rule-Based AI Chatbot in Python.

```python
print("Chatbot started. Type 'bye' to exit.")

while True:
    user = input("You: ").lower().strip()

    if user == "hello" or user == "hi":
        print("Bot: Hello! Welcome.")
    elif user == "how are you":
        print("Bot: I'm fine, thank you!")
    elif "your name" in user:
        print("Bot: I'm a simple rule-based chatbot.")
    elif "help" in user:
        print("Bot: Try saying hello, how are you, or bye.")
    elif user == "bye":
        print("Bot: Goodbye!")
        break
    else:
        print("Bot: Sorry, I don't understand that.")
```

```
Chatbot started. Type 'bye' to exit.
You:  hi
Bot: Hello! Welcome.
You:  how are you
Bot: I'm fine, thank you!
You:  bye
Bot: Goodbye!
```